

Project Documentation

1. Introduction

Project Title: Smart Library Request Workflow in ServiceNow

Team Members:

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2. Project Overview

Purpose:

The Smart Library Request Workflow is designed to streamline library book requests and approvals within ServiceNow. Users can request books, track status updates, and receive notifications, while librarians can manage approvals efficiently. The system also provides dashboards and reports for administrators to monitor book requests and user activity.

Features:

- User login with role-based access (User & Librarian)
- Borrow request submission with validation
- Approval and rejection workflow for librarians
- Email and system notifications for request status updates
- Dashboard displaying total requests, requests by user, and pending approvals
- Monthly and real-time reporting on borrow requests

3. Architecture

Frontend:

- Built with React.js using functional components and hooks
- State management with Redux
- Routing via React Router
- Axios used for API calls to the backend

Backend:

- Node.js with Express.js framework
- RESTful API endpoints for CRUD operations on users, books, and borrow requests
- Middleware for authentication, authorization, and error handling

Database:

- MongoDB collections for Users, Books, and Borrow Requests
- Mongoose used for schema definition and database interaction
- Borrow Requests reference User and Book collections for data integrity

4. Setup Instructions

Prerequisites:

- Node.js (v18+)
- MongoDB (local or Atlas cluster)
- npm or yarn package manager

Installation:

- **Clone the repository:** <https://github.com/narmadha271/smart-library-request-workflow-in--servicenow.git>
- **Install dependencies:**
cd client
npm install
cd ../server
npm install
- **Configure environment variables in .env file:**
PORT=5000
MONGO_URI=[MongoDB URI]
JWT_SECRET=[JWT Secret]

5. Folder Structure

Client (React Frontend):

```
client/  
├── public/  
├── src/  
│   ├── components/  
│   ├── pages/  
│   ├── redux/  
│   ├── services/  
│   ├── App.js  
│   └── index.js
```

Server (Node.js Backend):

```
server/
```

```
|— controllers/  
|— models/  
|— routes/  
|— middleware/  
|— config/  
|— server.js
```

6. Running the Application

Frontend:

```
cd client  
npm start
```

Backend:

```
cd server  
npm start
```

Frontend runs on port 3000, backend on port 5000.

7. API Documentation

Endpoints:

- POST /api/auth/login – User login
- POST /api/auth/register – User registration
- GET /api/books – Retrieve all books
- POST /api/borrow-requests – Submit a borrow request
- PUT /api/borrow-requests/:id/approve – Approve borrow request (librarian)
- GET /api/reports/monthly – Monthly request summary

Example Response:

```
{  
  "success": true,  
  "data": [ ... ]  
}
```

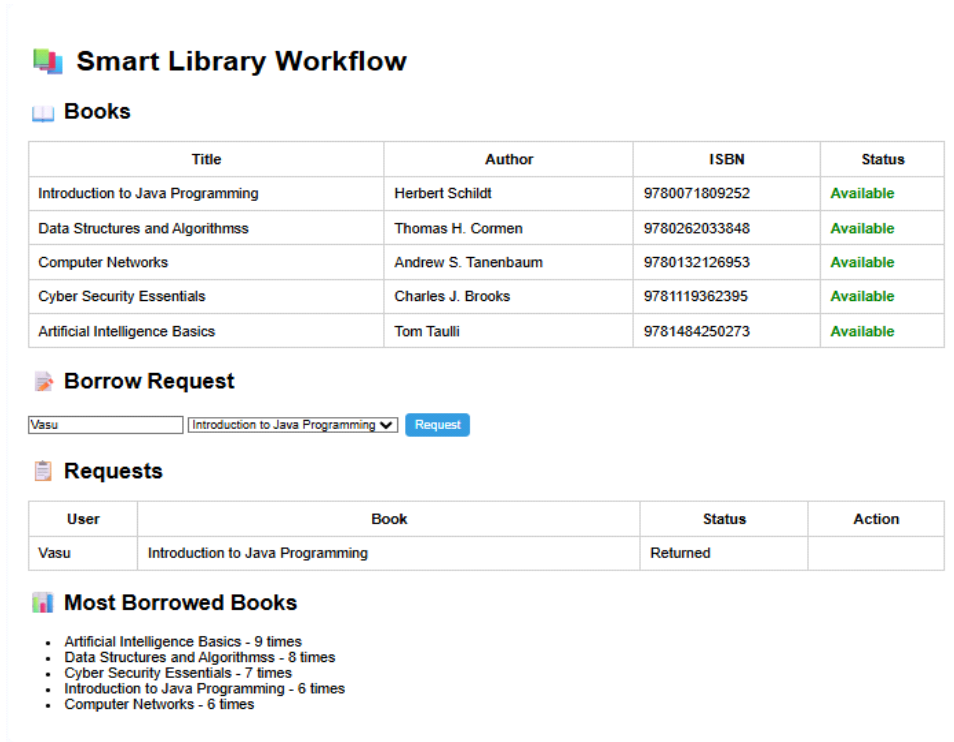
8. Authentication

- JWT-based authentication and authorization
- Tokens stored in local storage and sent in headers
- Role-based access control: User vs Librarian

- Middleware checks token validity and user roles for protected routes

9. User Interface

- Screenshots or GIFs showing:
 - Login Page



Smart Library Workflow

Books

Title	Author	ISBN	Status
Introduction to Java Programming	Herbert Schildt	9780071809252	Available
Data Structures and Algorithmss	Thomas H. Cormen	9780262033848	Available
Computer Networks	Andrew S. Tanenbaum	9780132126953	Available
Cyber Security Essentials	Charles J. Brooks	9781119362395	Available
Artificial Intelligence Basics	Tom Taulli	9781484250273	Available

Borrow Request

Vasu Introduction to Java Programming Request

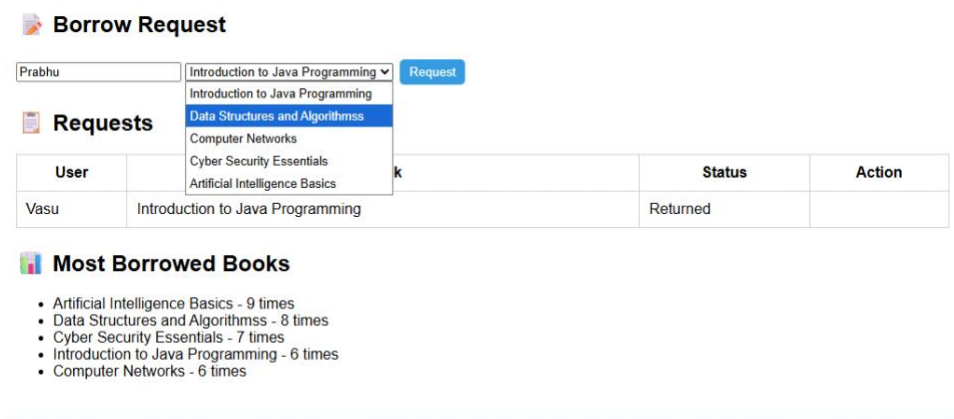
Requests

User	Book	Status	Action
Vasu	Introduction to Java Programming	Returned	

Most Borrowed Books

- Artificial Intelligence Basics - 9 times
- Data Structures and Algorithmss - 8 times
- Cyber Security Essentials - 7 times
- Introduction to Java Programming - 6 times
- Computer Networks - 6 times

- Borrow Request Form



Borrow Request

Prabhu Introduction to Java Programming Request

Requests

User	Book	Status	Action
Vasu	Introduction to Java Programming	Returned	

Most Borrowed Books

- Artificial Intelligence Basics - 9 times
- Data Structures and Algorithmss - 8 times
- Cyber Security Essentials - 7 times
- Introduction to Java Programming - 6 times
- Computer Networks - 6 times

- Approval Dashboard

Borrow Request

Requests

User	Book	Status	Action
Vasu	Introduction to Java Programming	Returned	
Prabhu	Cyber Security Essentials	Requested	Approve Reject

Most Borrowed Books

- Artificial Intelligence Basics - 9 times
- Data Structures and Algorithmss - 8 times
- Cyber Security Essentials - 7 times
- Introduction to Java Programming - 6 times
- Computer Networks - 6 times

- Reports and Charts

Requests

User	Book	Status	Action
Vasu	Introduction to Java Programming	Returned	
Prabhu	Cyber Security Essentials	Approved	Return Book
Gopi	Data Structures and Algorithmss	Approved	Return Book
Melki	Introduction to Java Programming	Approved	Return Book
Almahin	Computer Networks	Approved	Return Book
Sathees	Artificial Intelligence Basics	Approved	Return Book

Most Borrowed Books

- Artificial Intelligence Basics - 10 times
- Data Structures and Algorithmss - 9 times
- Cyber Security Essentials - 8 times
- Introduction to Java Programming - 7 times
- Computer Networks - 7 times

10. Testing

- Backend unit testing using Jest
- API integration testing
- Manual UAT performed for all user workflows

11. Screenshots or Demo

- Demo link: <https://drive.google.com/file/d/1xLN-mOIaJvgr9j-1GmgazcKD2e6Z3Dtu/view?usp=sharing>

12. Known Issues

- Dashboard may experience slight delays with large datasets
- Minor UI inconsistencies on mobile devices

13. Future Enhancements

- Add search and filter functionality for books
- Email notifications for overdue or unreturned books
- Analytics for user activity trends
- Improved mobile responsiveness